Computer Vision

Radiometry

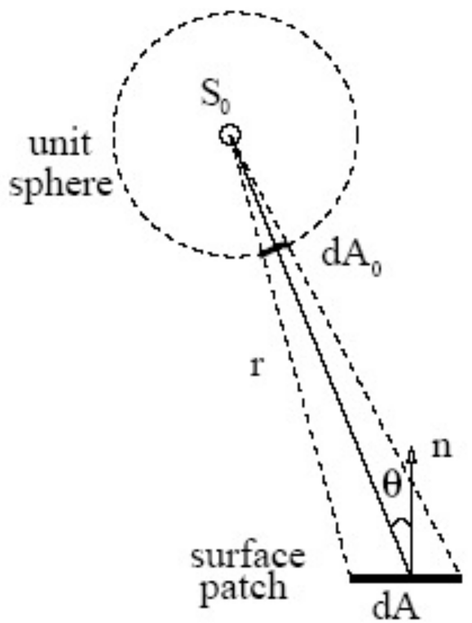
Darius Burschka

Radiometry

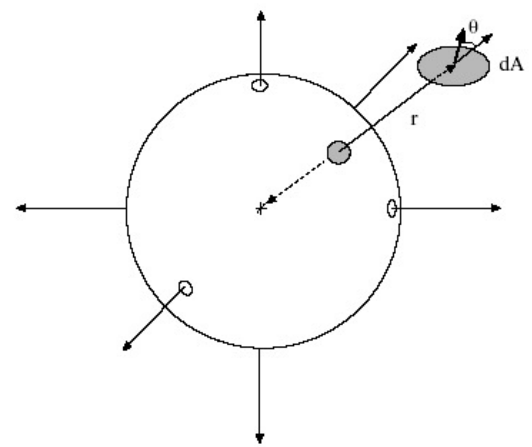
- *Radiometry* is the part of image formation concerned with the relation among the amounts of
  - light energy emitted from light sources,
  - reflected from surfaces,
  - and registered by sensors.

Solid Angle

- *Solid angle* is defined by <u>the projected area of a surface patch onto a unit sphere of a point</u>.



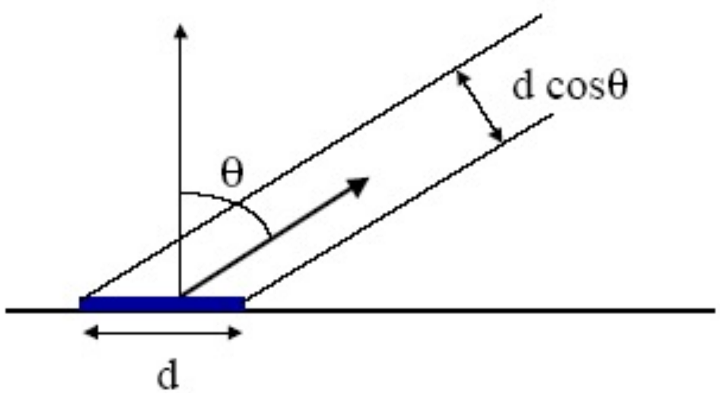
$$d\omega = dA_0 = \frac{dA\cos\theta}{r^2}$$



(Solid angle is subtended by a point and a surface patch.)

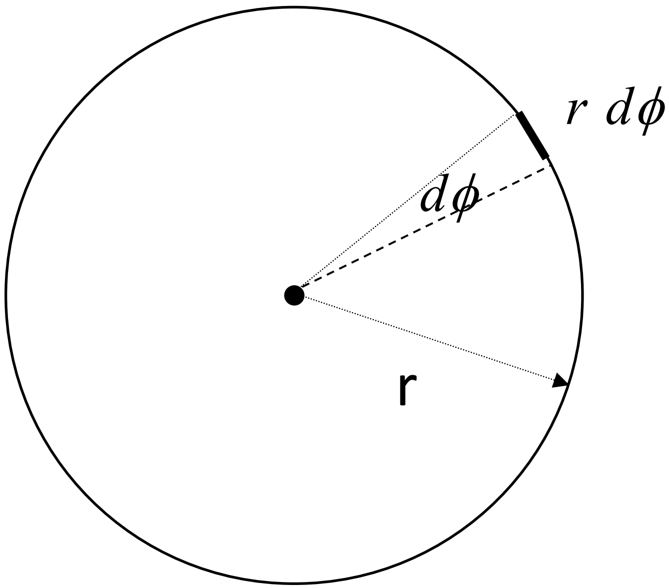
Foreshortening

- A big source, viewed at a glancing angle, must produce the same effect as a small source viewed frontally.
- This phenomenon is known as *foreshortening*.



Solid Angle

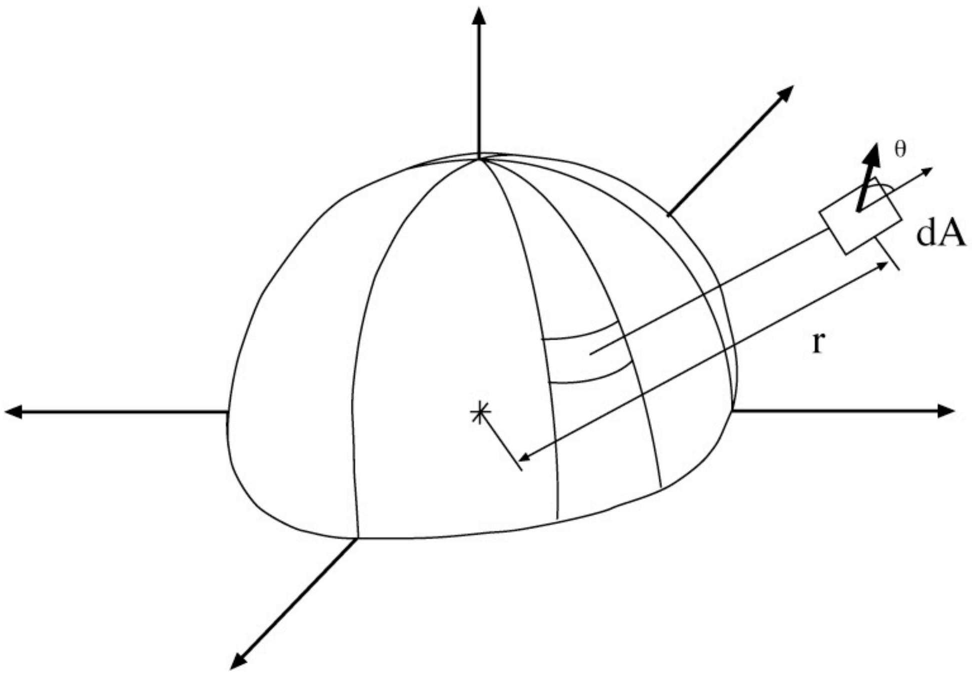
- Arc length



Solid Angle

Defined by analogy with angle (in radians)

The solid angle subtended by an infinitesimal patch area dA is given by



$$d\omega = \sin\vartheta (d\vartheta)(d\phi)$$

Another useful expression:

Radiometry

Core Questions:

how “bright” will surfaces be?

what is “brightness”?

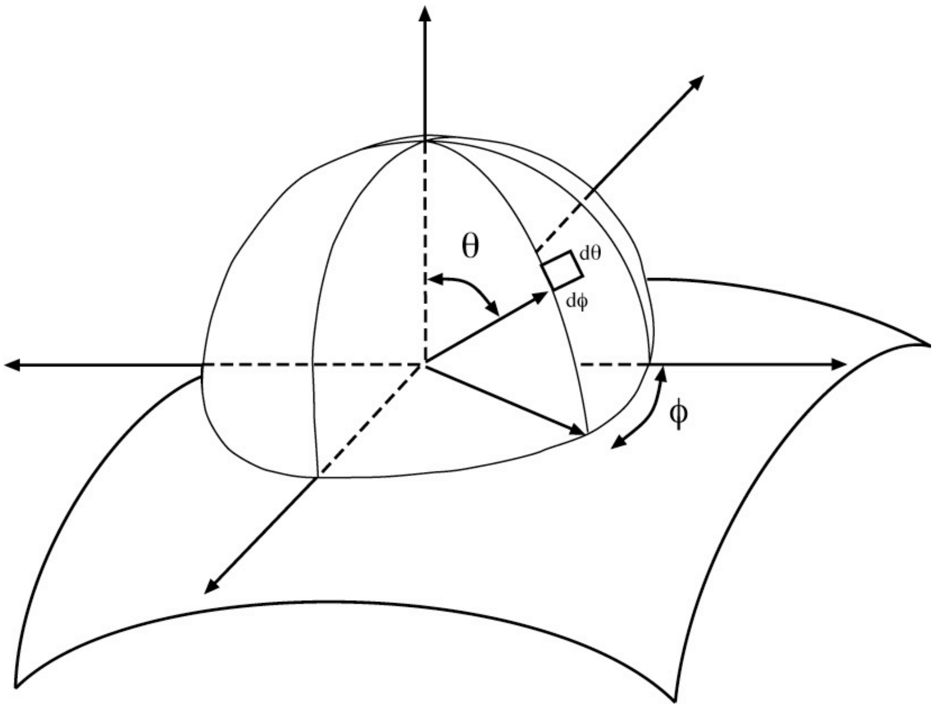
measuring light

interactions between light and surfaces

Core idea - think about light arriving at a surface

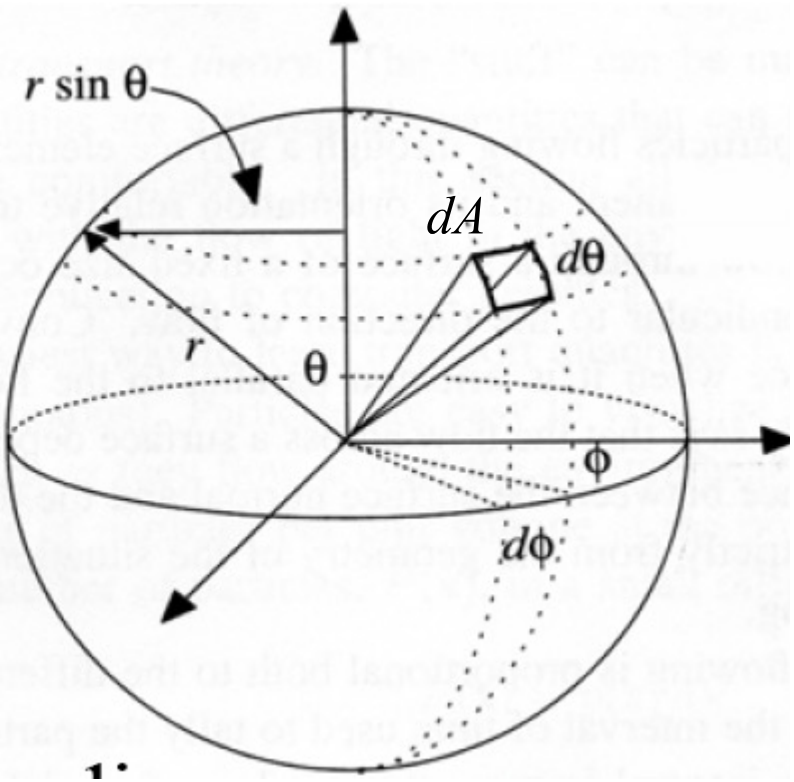
around any point is a hemisphere of directions

Simplest problems can be dealt with by reasoning about this hemisphere



Solid Angle

- *Solid angle* is defined by the projected area of a surface patch onto a unit sphere of a point.



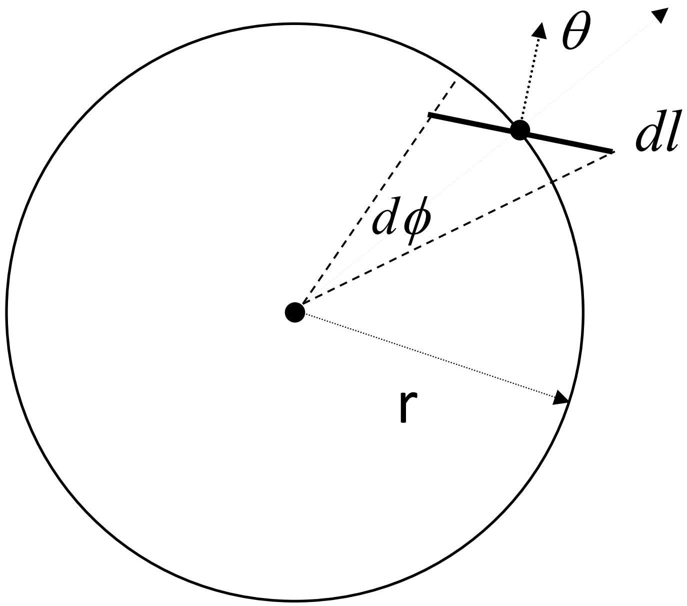
$$dA = rd\theta r\sin\theta d\phi = r^2\sin\theta d\theta d\phi$$

$$TotalArea = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} r^2\sin\theta d\theta d\phi = 4\pi r^2$$

$$dw = \frac{dA}{r^2} = \sin\theta d\theta d\phi$$

Solid Angle

- Similarly, solid angle due to a line segment is



$$d\phi = \frac{dl \cos\theta}{r}$$

Radiance

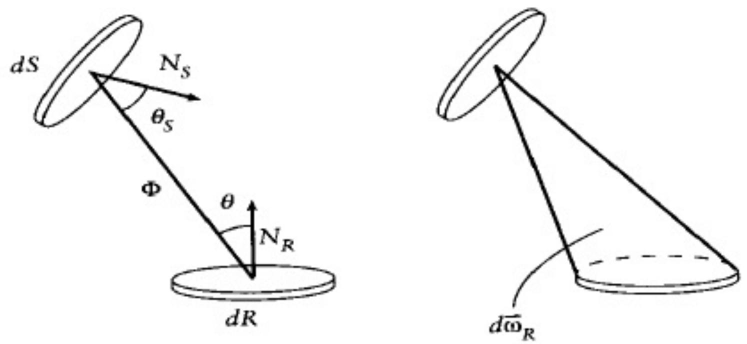
- The distribution of light in space is a function of position and direction.
- The appropriate unit for measuring the distribution of light in space is *radiance*, which is defined as the *power* (the amount of energy per unit time) traveling at some point in a specified direction, per unit area *perpendicular to the direction of travel*, per unit solid angle.
- In short, radiance is the amount of light radiated from a point… (into a unit solid angle, from a unit area).

Radiance = Power / (solid angle x foreshortened area)

W/sr/m2
W is Watt, sr is steradian, m2 is meter-squared

Radiance

- Radiance from dS to dR

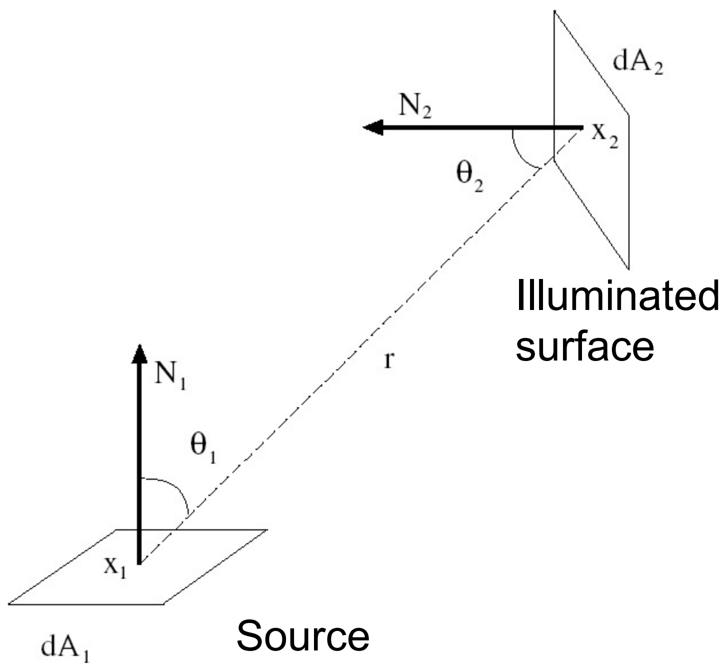


Radiance = Power / (solid angle x foreshortened area)

Radiance

- Example: Infinitesimal source and surface patches

Radiance = Power / (solid angle x foreshortened area)



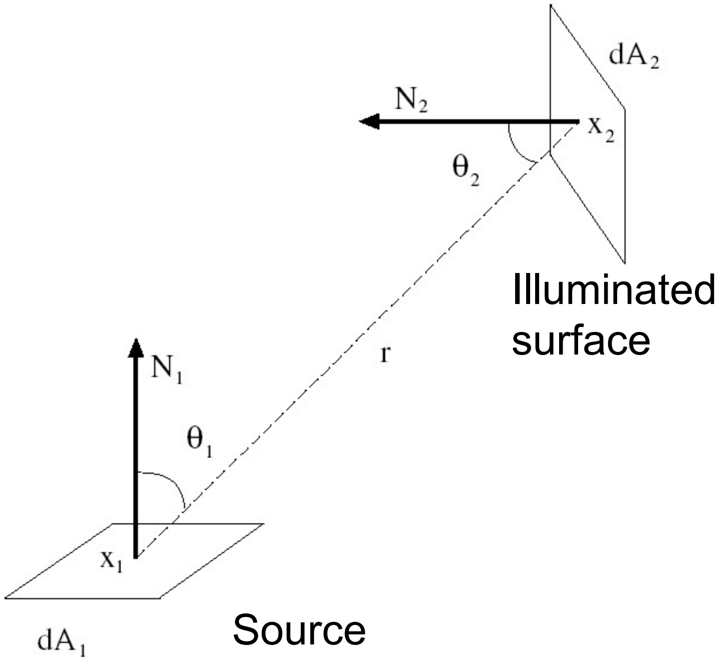
Radiance at x1 leaving to x2

$$L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2) = \frac{d\psi}{dw \cos\theta_1 dA_1} = \frac{r^2 d\psi}{dA_2 \cos\theta_2 \cos\theta_1 dA_1}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Radiance

Radiance = Power / (solid angle x foreshortened area)



Power at x1 leaving to x2

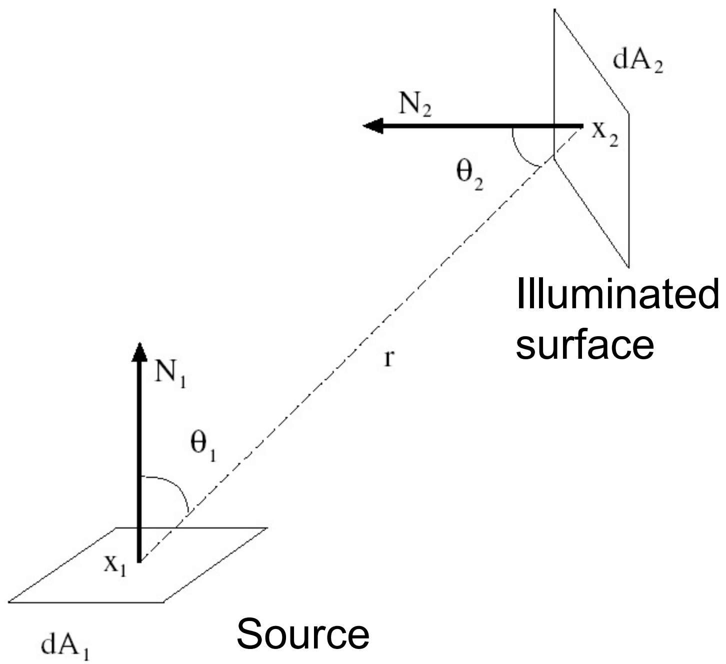
$$d\psi = L(\mathbf{x}_1, \mathbf{x}_1 \to \mathbf{x}_2)\, dw \cos\theta_1 dA_1$$

$$= \frac{L(\mathbf{x}_1, \mathbf{x}_1 \to \mathbf{x}_2)\, dA_2 \cos\theta_2 \cos\theta_1 dA_1}{r^2}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Radiance

- The medium is vacuum, that is, it does not absorb energy. Therefore, the power reaching point x2 is equal to the power leaving for x2 from x1.



Power at x2 from direction x1 is

$$d\psi = \frac{L(\mathbf{x}_1, \mathbf{x}_1 \to \mathbf{x}_2) dA_2 \cos\theta_2 \cos\theta_1 dA_1}{r^2}$$

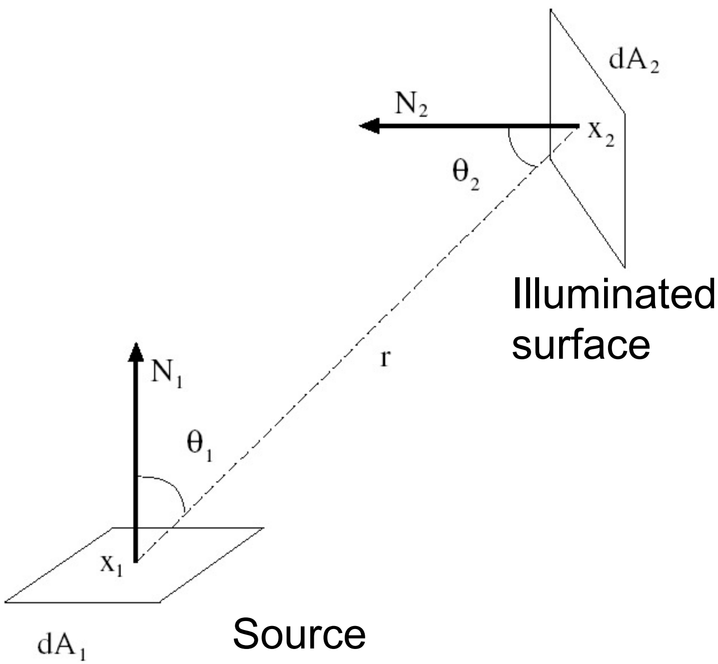
Let the radiance arriving at x2 from the direction of x1 is

$$L(\mathbf{x}_2, \mathbf{x}_1 \to \mathbf{x}_2) = \frac{d\psi}{dw \cos\theta_2 dA_2} = \frac{r^2 d\psi}{dA_1 \cos\theta_1 \cos\theta_2 dA_2}$$

$$dw = \frac{dA_1 \cos\theta_1}{r^2}$$

Radiance

- **Radiance is constant along a straight line.**



$$L(\mathbf{x}_1, \mathbf{x}_1 \to \mathbf{x}_2) = L(\mathbf{x}_2, \mathbf{x}_1 \to \mathbf{x}_2)$$

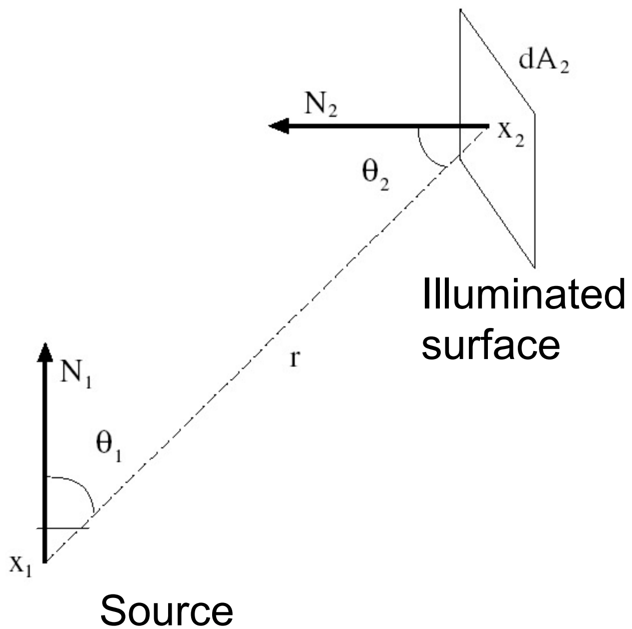
Point Source

- Many light sources are physically small compared with the environment in which they stand.
- Such a light source is approximated as an extremely small sphere, in fact, a point.
- Such a light source is known as a *point source*.

Radiance Intensity

- If the source is a *point source*, we use radiance intensity.

Radiance intensity = Power / (solid angle)



$$I = \frac{d\psi}{dw} = \frac{r^2 d\psi}{dA_2 \cos\theta_2}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Light at Surfaces

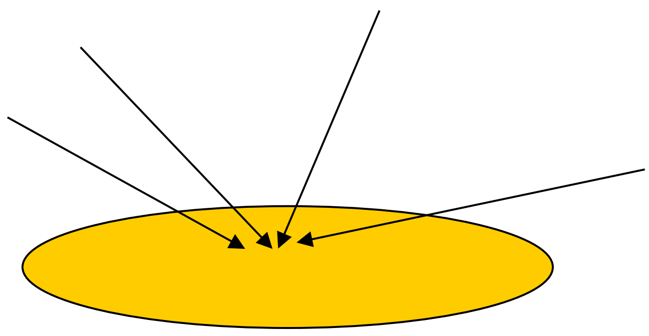
- When light strikes a surface, it may be absorbed, transmitted, or scattered; usually, combination of these effects occur.
- It is common to assume that all effects are local and can be explained with a **local interaction model**. In this model:
  - The radiance leaving a point on a surface is due only to radiance arriving at this point.
  - Surfaces do not generate light internally and treat sources separately.
  - Light leaving a surface at a given wavelength is due to light arriving at that wavelength.

Light at Surfaces

- In the local interaction model, *fluorescence*, [absorb light at one wavelength and then radiate light at a different wavelength], and *emission* [e.g., warm surfaces emits light in the visible range] are neglected.

Irradiance

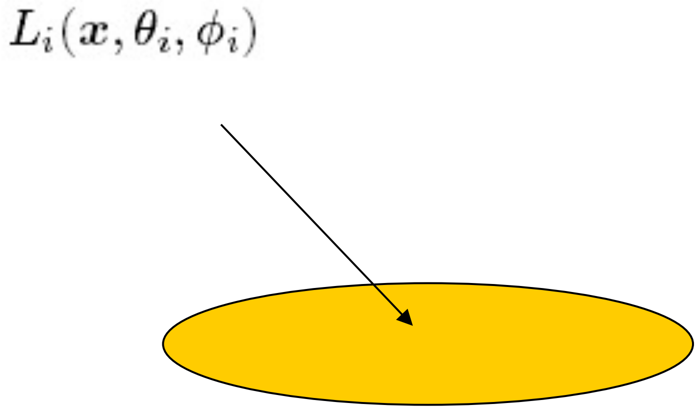
- *Irradiance* is the total incident power per unit area.



Irradiance = Power / Area

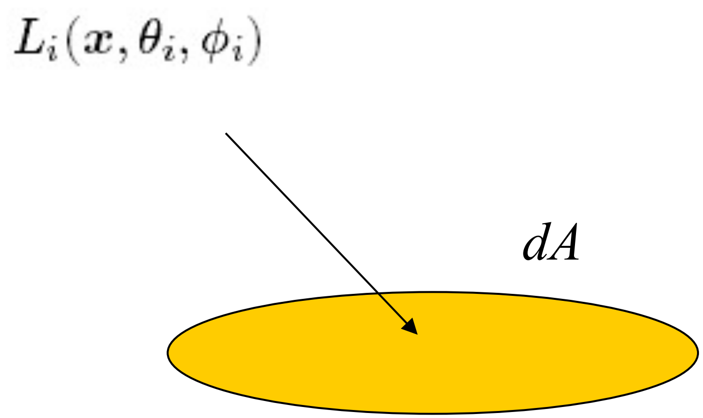
Irradiance

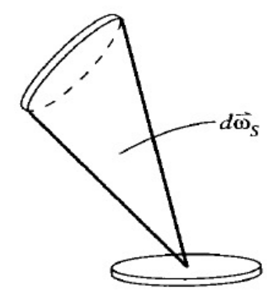
- What is the irradiance due to source from angle (θ_i, ϕ_i)?



Irradiance

- What is the irradiance due to source from angle (θ_i, ϕ_i) ?

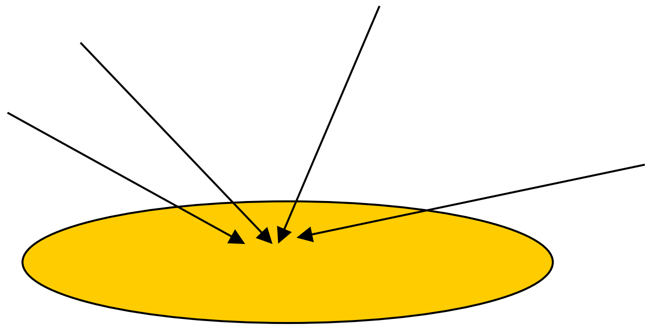




$$Irradiance = \frac{d\psi}{dA} = \frac{L_i(x,\theta_i,\phi_i)dw\cos\theta_i dA}{dA} = L_i(x,\theta_i,\phi_i)dw\cos\theta_i$$

Irradiance

- What is the total irradiance?



Integrate over the whole hemisphere.

Exercise: Suppose the radiance is constant from all directions. Calculate the irradiance.

Irradiance due to a Point Source

- For a point source,

$$\texttt{Radiance\_Intensity} = \frac{\texttt{power}}{\texttt{solid\_angle\_source}}$$

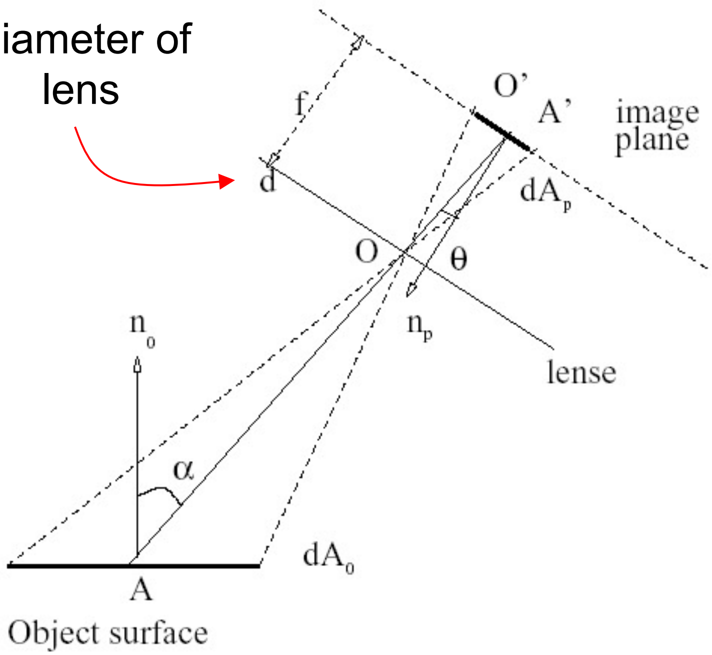
$$I = \frac{d\psi}{dw} = \frac{r^2 d\psi}{dA_i \cos\theta_i}$$

$$dw = \frac{dA_i \cos\theta_i}{r^2}$$

$$\Rightarrow \quad d\psi = I \frac{dA_i \cos\theta_i}{r^2} \quad \Rightarrow \quad Irradiance = \frac{d\psi}{dA_i} = I \frac{\cos\theta_i}{r^2}$$

The Relationship Between Image Intensity and Object Radiance

Diameter of lens



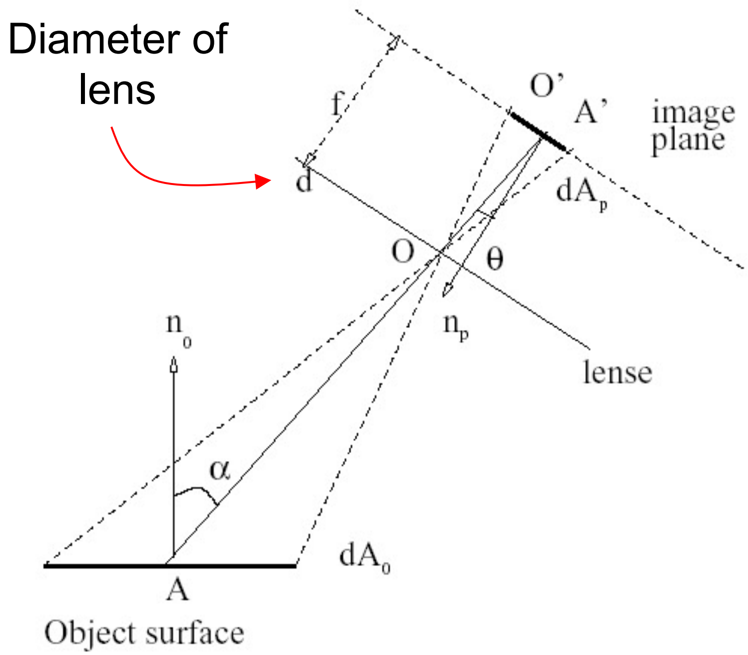
We assume that there is no power loss in the lens.

The power emitted to the lens is

$$d\psi = L_{object} dA_0 \cos\alpha \, dw_0$$

Radiance of object

The Relationship Between Image Intensity and Object Radiance



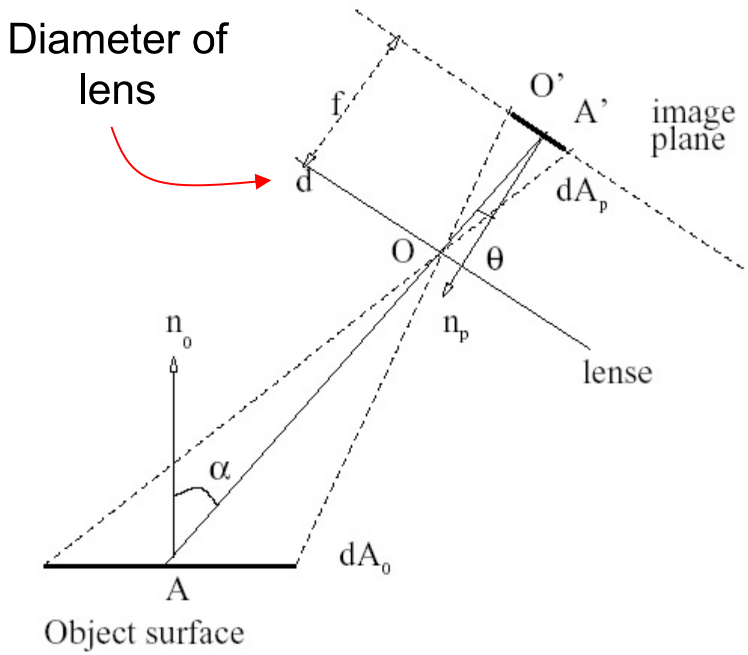
The solid angle for the entire lens is

$$dw_0 = \frac{(\pi d^2/4)\cos\theta}{r^2}$$

The power emitted to the lens is

$$d\psi = L_{object} dA_0 \cos\alpha \, dw_0$$
$$= L_{object} dA_0 \cos\alpha \frac{\pi d^2 \cos\theta}{4r^2}$$

The Relationship Between Image Intensity and Object Radiance



The solid angle at O can be written in two ways.

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos\theta}{|OA'|^2}$$

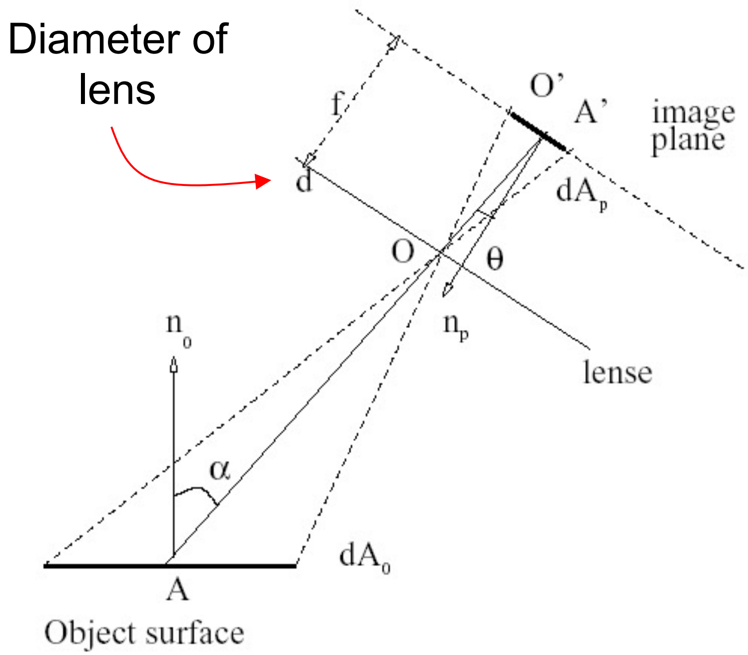
Note that

$$|OA'| = f / \cos\theta$$

Therefore

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos^3\theta}{f^2}$$

The Relationship Between Image Intensity and Object Radiance



Combine

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos^3\theta}{f^2}$$

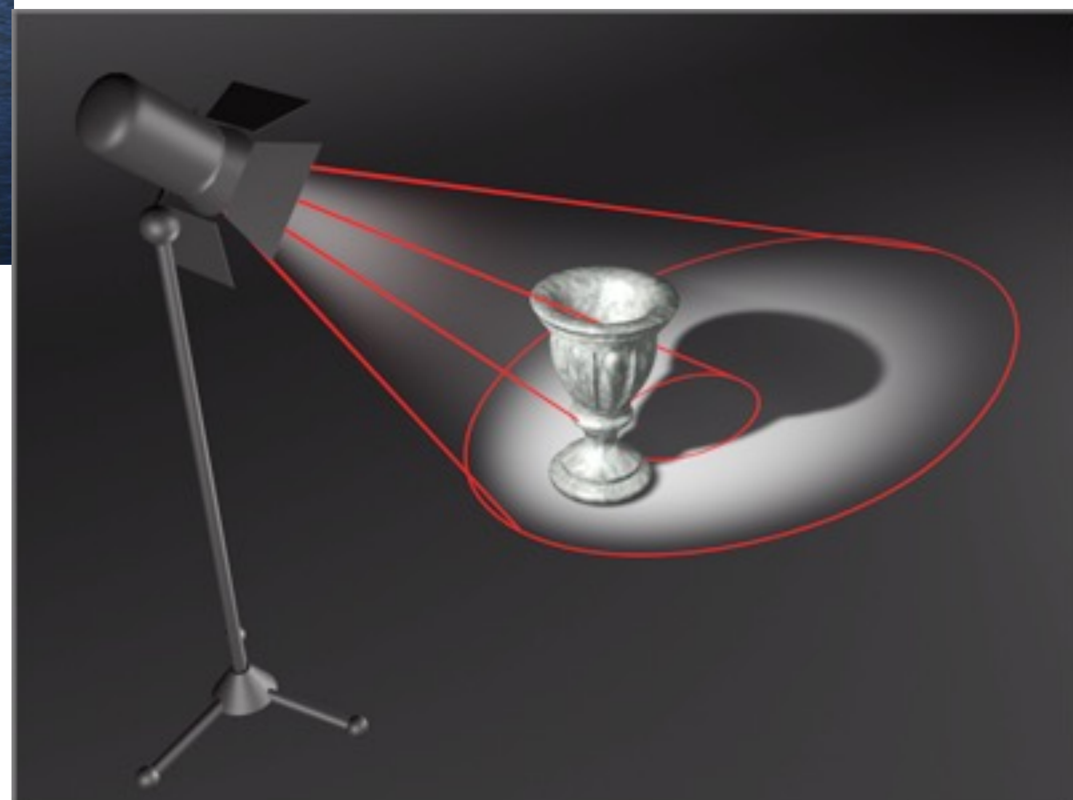
$$d\psi = L_{object} dA_0 \cos\alpha \frac{\pi d^2 \cos\theta}{4r^2}$$

to get

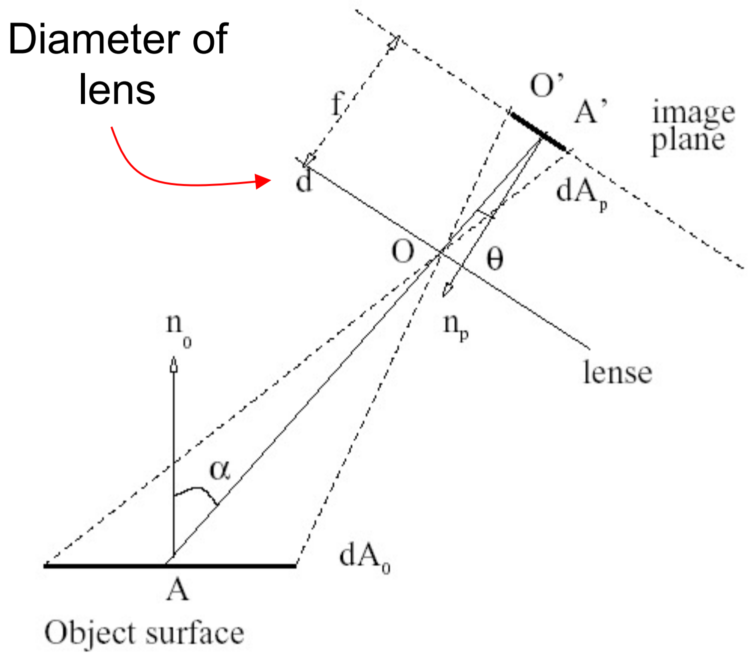
$$d\psi = \left(\frac{\pi}{4}\right) L_{object} \left(\frac{d}{f}\right)^2 \cos^4\theta \, dA_p$$

Light Falloff

$$d\psi = \left(\frac{\pi}{4}\right) L_{object} \left(\frac{d}{f}\right)^2 \cos^4 \theta dA_p$$



The Relationship Between Image Intensity and Object Radiance

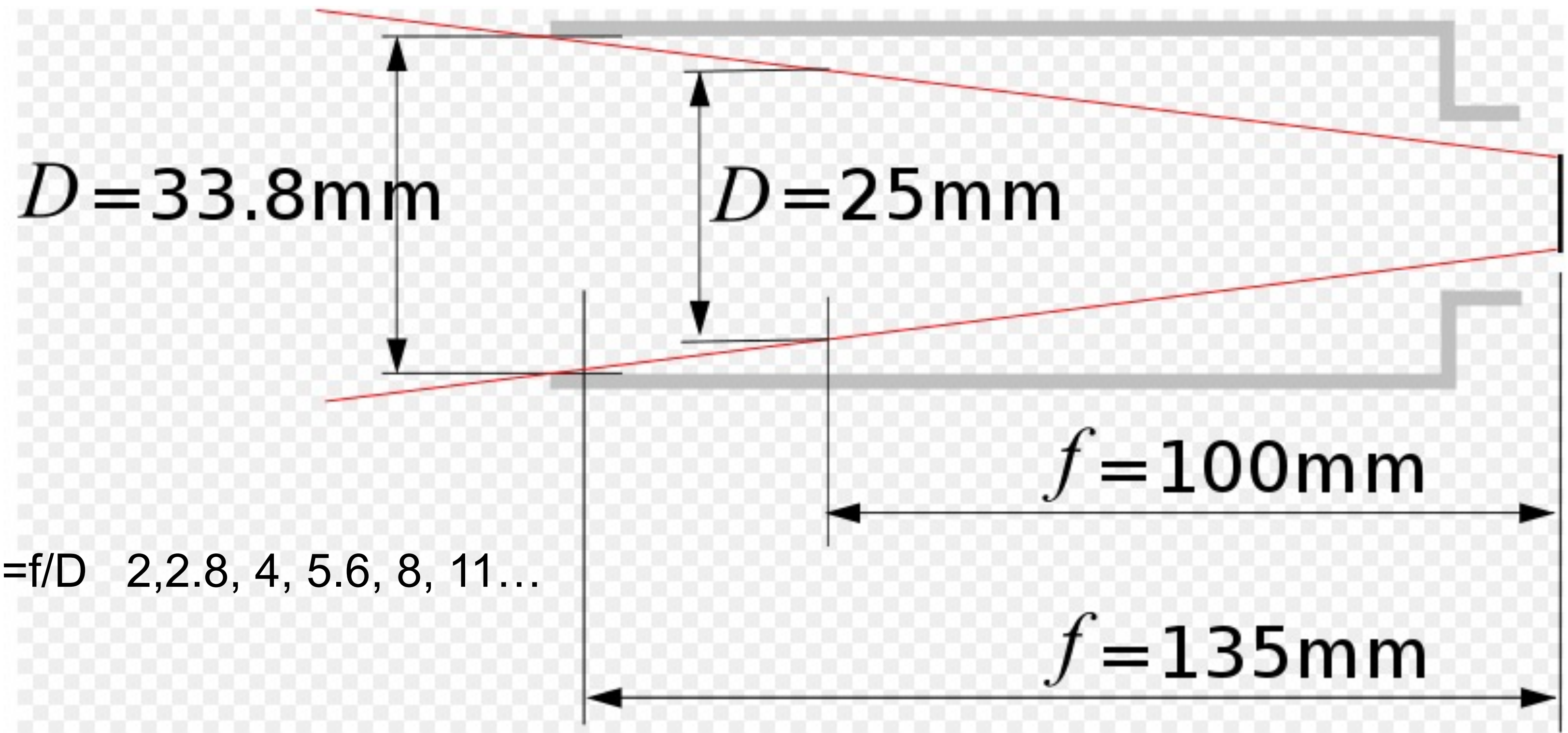


Therefore the irradiance on the image plane is

$$Irradiance = \frac{d\psi}{dA_p} = \left(\frac{\pi}{4}\right) L_{object} \left(\frac{d}{f}\right)^2 \cos^4\theta$$

The irradiance is converted to pixel intensities, which is directly proportional to the radiance of the object.

Different Focal Lengths Same Illuminanace (F-number)



F=f/D   2,2.8, 4, 5.6, 8, 11…

F-Number

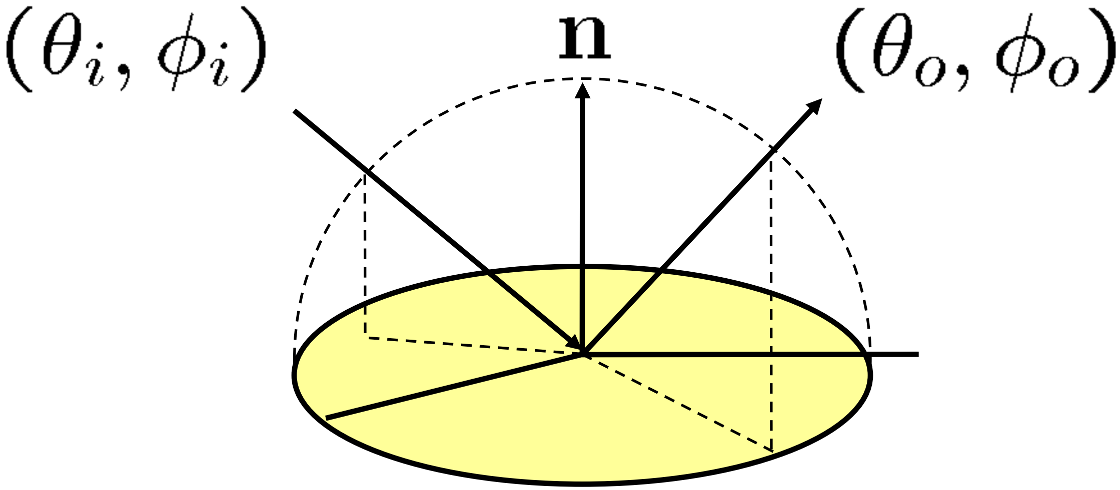
Depth of field increases with f-number, as illustrated in the photos to the right. This means that photos taken with a low f-number will tend to have one subject in focus, with the rest of the image out of focus. This is frequently useful for nature photography or certain special effects.

f/32

f/5

Surface Characteristics

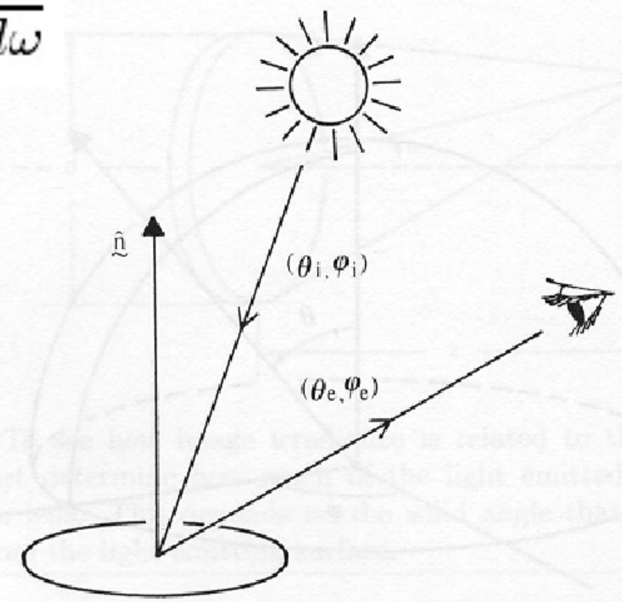
- We want to describe the relationship between incoming light and reflected light.
- This is a function of both the direction in which light arrives at a surface and the direction in which it leaves.



Bidirectional Reflectance Distribution Function (BRDF)

- BRDF is defined as the ratio of the radiance in the outgoing direction to the incident irradiance.

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(\boldsymbol{x}, \theta_o, \phi_o)}{L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos \theta_i d\omega}$$



Horn, 1986

Figure 10-7. The bidirectional reflectance distribution function is the ratio of the radiance of the surface patch as viewed from the direction (θ_e, ϕ_e) to the irradiance resulting from illumination from the direction (θ_i, ϕ_i).

Bidirectional Reflectance Distribution Function (BRDF)

- The radiance leaving a surface due to irradiance *in a particular direction* is easily obtained from the definition of BRDF:

$$L_o(x, \theta_o, \phi_o) = \rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) L_i(x, \theta_i, \phi_i) \cos\theta_i d\omega$$

Bidirectional Reflectance Distribution Function (BRDF)

- The radiance leaving a surface due to irradiance *in all incoming directions* is

$$L_o(\boldsymbol{x}, \theta_o, \phi_o) = \int_\Omega \rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos\theta_i d\omega$$

where Omega is the incoming hemisphere.

Lambertian Surface

- A *Lambertian surface* has constant BRDF.

constant

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(\boldsymbol{x}, \theta_o, \phi_o)}{L_i(\boldsymbol{x}, \theta_i, \phi_i)\cos\theta_i d\omega}$$

Lambertian Surface

- A Lambertian surface looks equally bright from any view direction.
- The image intensities of the surface only changes with the illumination directions.

constant

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(\boldsymbol{x}, \theta_o, \phi_o)}{L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos \theta_i d\omega}$$

Lambertian Surface

- For a Lambertian surface, the outgoing radiance is proportional to the incident radiance.

constant

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(\boldsymbol{x}, \theta_o, \phi_o)}{L_i(\boldsymbol{x}, \theta_i, \phi_i)\cos\theta_i d\omega}$$

- If the light source is a point source, a pixel intensity will only be a function of $cos\theta_i$

Remember, for a point source $Irradiance = \dfrac{d\psi}{dA_i} = I\dfrac{\cos\theta_i}{r^2}$

Point source

BRDF is a constant. These surfaces look equally bright from all viewing directions.

$$f(\theta_i, \phi_i, \theta_e, \phi_e) = \frac{1}{\pi}$$

Radiance reflected from Lambertian surface illuminated by point source:

$$L(\theta_e, \phi_e) = \int_{-\pi}^{\pi} \int_{0}^{\pi/2} \frac{1}{\pi} \delta(\theta_i - \theta_0) \delta(\phi_i - \phi_0) \sin(\theta_i) \cos(\theta_i) \, d\theta_i \, d\phi_i$$

$$\propto \cos(\theta_0)$$

Lambertian surfaces and albedo

for a Lambertian surface, BRDF is independent of angle.

Useful fact:

$$\rho_{brdf} = \frac{\rho_d}{\pi}$$

$$L_o(\underline{x}) = \rho_{brdf} L_i(\underline{x}, \vartheta_i, \varphi_i) \cos \vartheta_i d\omega$$

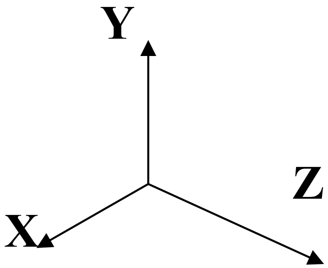
Suppose now that we have an idea pinhole source in a given direction. Then we if look at the entire power

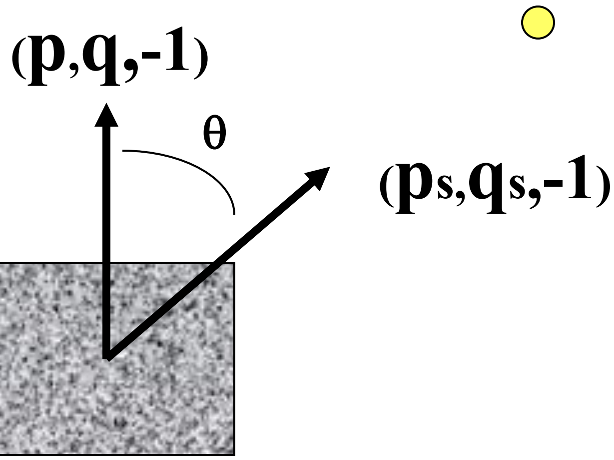
$$L_o(\underline{x}) = \rho_{brdf} \int_{\Omega} L_i(\underline{x}, \vartheta_i, \varphi_i) \cos \vartheta_i d\omega = \rho_{brdf} L_{source} \cos \vartheta_{source}$$

LAMBERTIAN REFLECTANCE MAP

LAMBERTIAN MODEL

$$E = L \rho \, COS\,\theta$$





$$COS\theta = \frac{1 + pp_s + qq_s}{\sqrt{1 + p^2 + q^2}\sqrt{1 + p_s^{\ 2} + q_s^{\ 2}}}$$

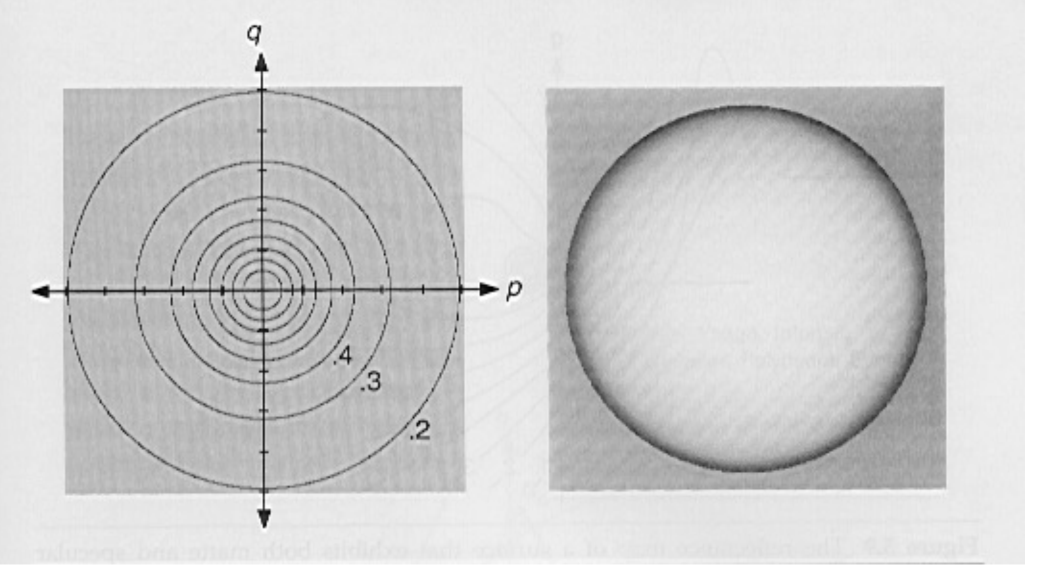
LAMBERTIAN REFLECTANCE MAP

$$E = L\rho \frac{1 + pp_s + qq_s}{\sqrt{1 + p^2 + q^2}\sqrt{1 + p_s{}^2 + q_s{}^2}}$$

Grouping L and ρ as a constant , local surface orientations that produce equivalent intensities under the Lambertian reflectance map are quadratic conic section contours in gradient space.

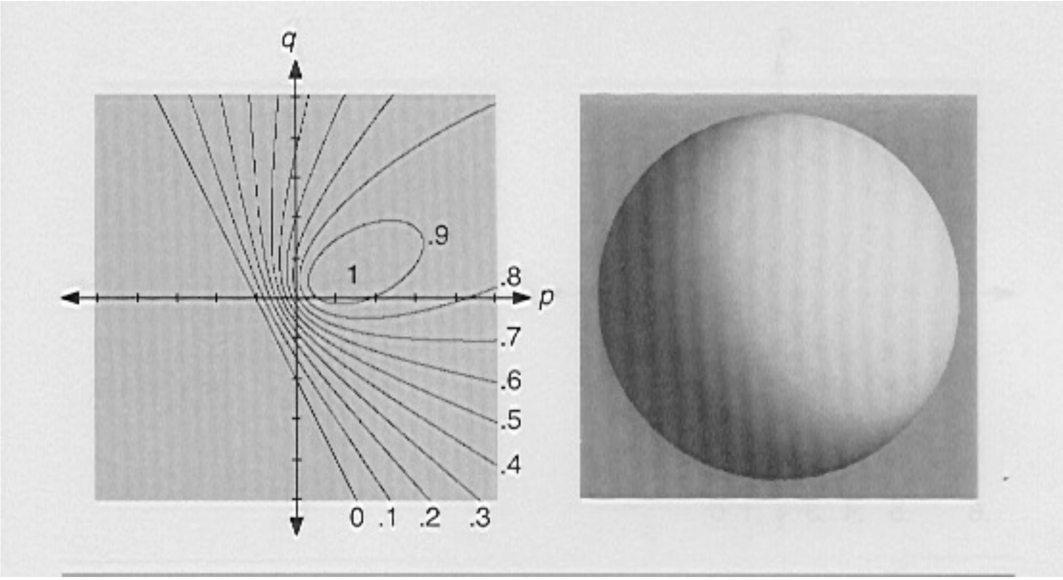
$$I = \frac{1 + pp_s + qq_s}{\sqrt{1 + p^2 + q^2}\sqrt{1 + p_s{}^2 + q_s{}^2}}$$

LAMBERTIAN REFLECTANCE MAP



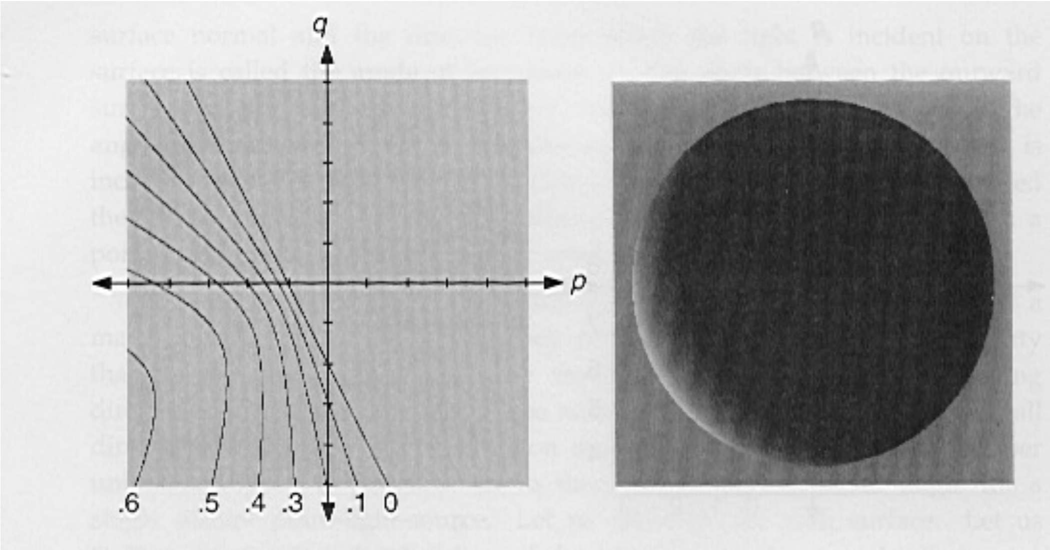
$p_s=0$ $q_s=0$

LAMBERTIAN REFLECTANCE MAP



p_s=0.7 q_s=0.3

LAMBERTIAN REFLECTANCE MAP



p_s= -2 q_s= -1

Linear shading map

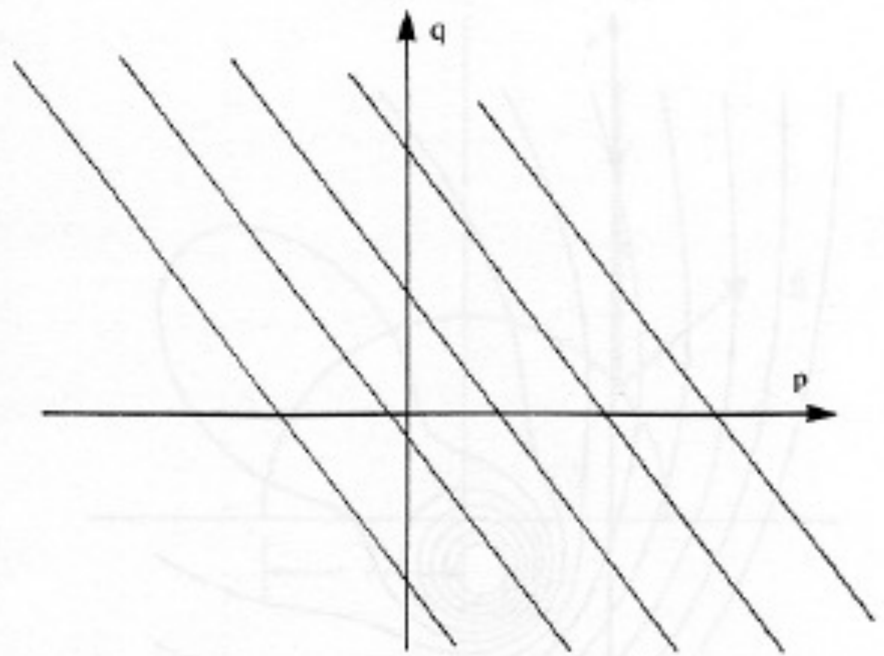


Figure 10-14. In the case of the material in the maria of the moon, the reflectance map can be closely approximated by a function of a linear combination of the components of the gradient. The contours of constant brightness are parallel straight lines in gradient space.

Horn, 1986

Linear shading: 1st order terms of Lambertian shading

Lambertian point source

$$R(p,q) = k\frac{1+p_s p+q_s q}{\sqrt{1+p^2+q^2}\sqrt{1+p_s^2+q_s^2}}$$

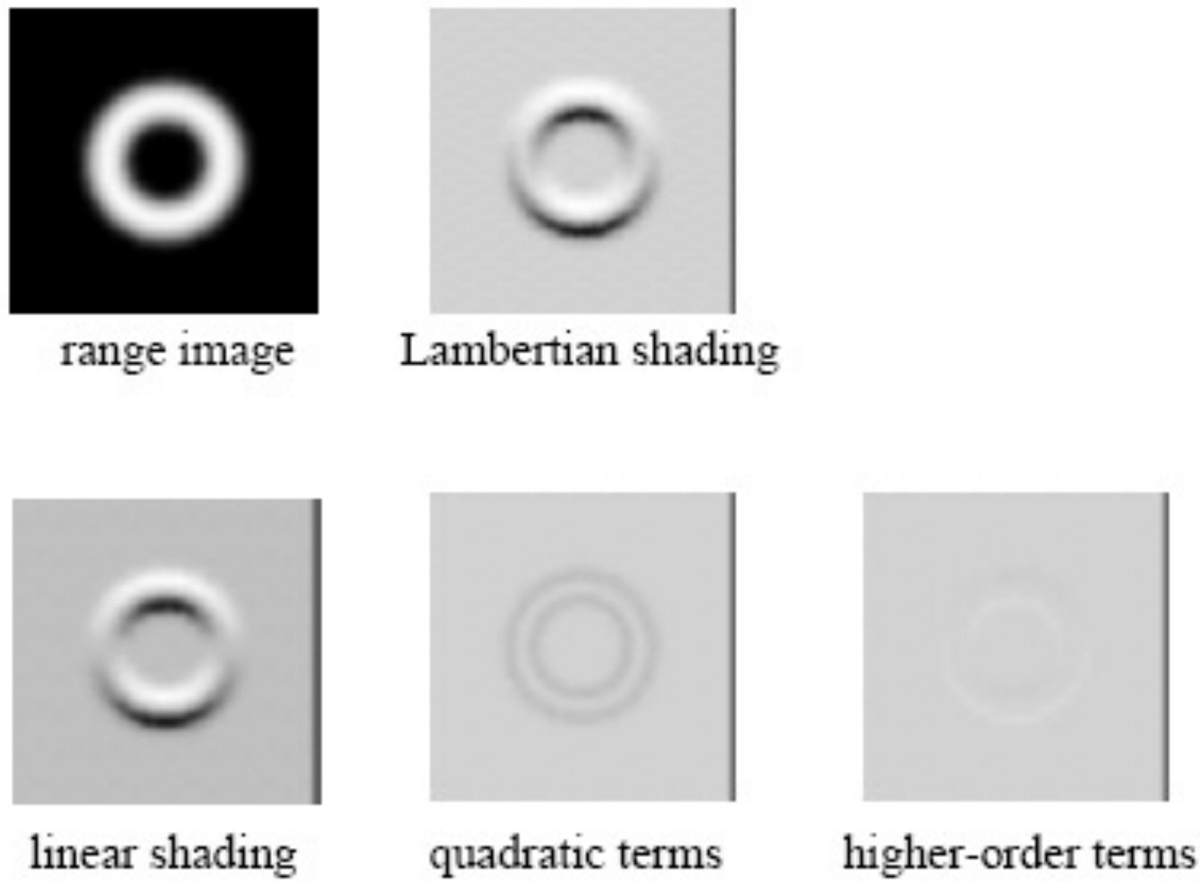
1st order Taylor series about p=q=0

$$\approx k_2 + \left.\frac{\partial R(p,q)}{\partial p}\right|_{p=0,q=0} p + \left.\frac{\partial R(p,q)}{\partial q}\right|_{p=0,q=0} q$$

$$= k_2(1+p_s p+q_s q)$$

See Pentland, IJCV vol. 1 no. 4, 1990.

Linear shading



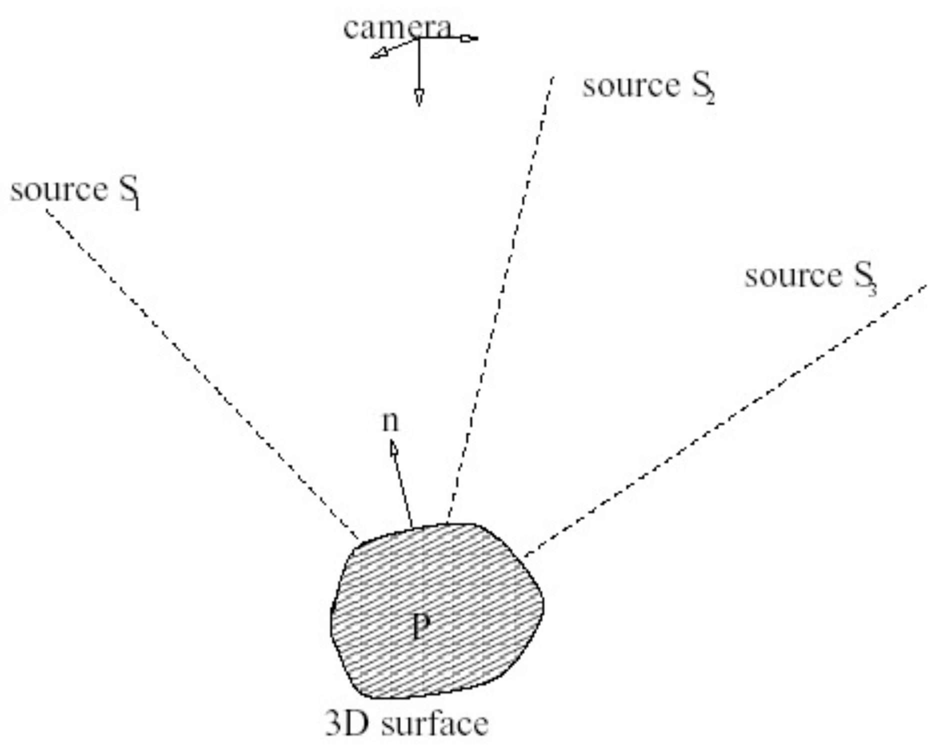
Pentland 1990, Adelson&Freeman, 1991

Advantages of linear shading

- Linear relationship between surface range map and rendered image.
- Rendering is easy: differentiate with respect to azimuthal light source direction.
- Applies: linear sources, or shallow illumination angles and Lambertian surface.
- Allows for very simple inverse transformation from rendered image to surface range map, which we'll discuss later with shape-from-shading material.

Photometric Stereo

- If we are given a set of images of the same scene taken under different given lighting sources, can we recover the 3D shape of the scene?



Photometric Stereo

- For a point source and a Lambertian surface, we can write the image intensity as

$$I = k_d cos\theta = k_d \mathbf{s}^T \mathbf{n}$$

- Suppose we are given the intensities under three lighting conditions:

$$I_i = k_d \mathbf{s}_i^T \mathbf{n},\ i = 1, 2, 3$$

Camera and object are fixed, so a particular pixel intensity is only a function of lighting direction s_i.

Photometric Stereo

- Stack the pixel intensities to get a vector

$$\mathbf{I} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = k_d \begin{bmatrix} \mathbf{s}_1^T \\ \mathbf{s}_2^T \\ \mathbf{s}_3^T \end{bmatrix} \mathbf{n} = k_d \mathbf{S} \mathbf{n}$$

- The surface normal can be found as

$$k_d \mathbf{n} = \mathbf{S}^{-1} \mathbf{I}$$

- Since **n** is a unit vector

$$\mathbf{n} = \frac{\mathbf{S}^{-1}\mathbf{I}}{||\mathbf{S}^{-1}\mathbf{I}||}$$

Photometric Stereo

- If we have more than three sources, we can find the least squares estimate using the pseudo inverse:

$$\mathrm{n} = \frac{\mathbf{S}^{\dagger}\mathbf{I}}{||\mathbf{S}^{\dagger}\mathbf{I}||}$$

- As a result, we can find the surface normal of each point, hence the 3D shape

frame 2

frame 11

. . .

. . .

Approach 1

- Photograph the object with a calibration object in the picture, or available in another photograph.
- Use the multiple responses of the calibration object to the different light sources to form a look-up table.
- Index into that table using the multiple responses of the unknown object.
- Handles arbitrary BRDF.

Approach 2

- Assume a particular functional form for the BRDF (Lambertian). Assume known light source positions (point sources at infinity as specified locations).
- Analytically determine the surface slope for each location's collection of image intensities.

Photometric stereo

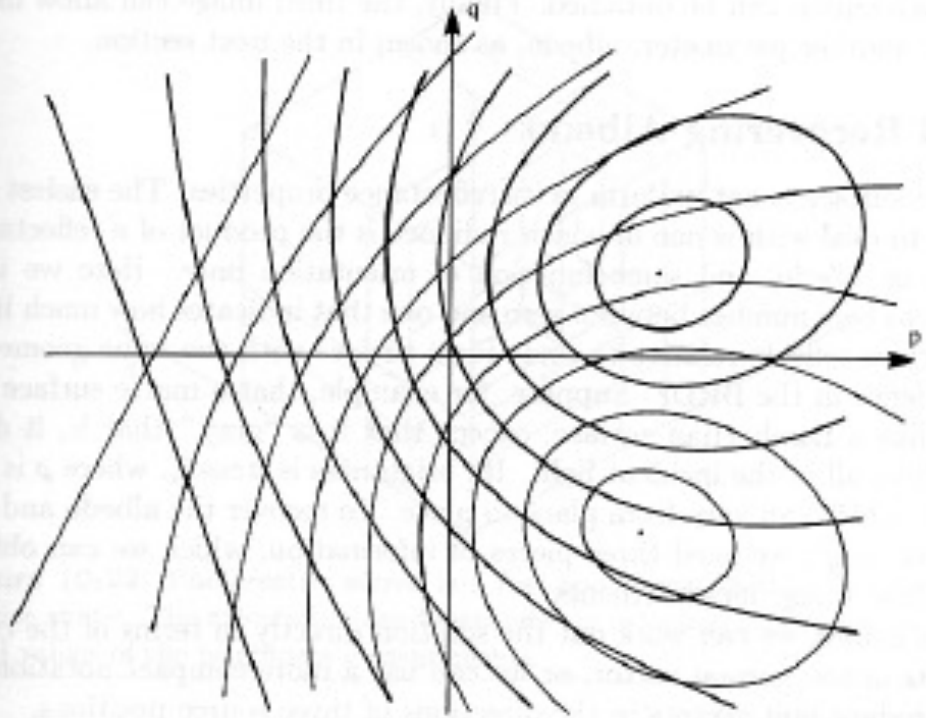


Figure 10-21. In the case of a Lambertian surface illuminated successively by two different point sources, there are at most two surface orientations that produce a particular pair of brightness values. These are found at the intersection of the corresponding contours in two superimposed reflectance maps.

See Forsyth&Ponce sect. 5.4 for procedure. In HW: don't need to integrate the surface normals to get the shape.

From the image under the ith lighting condition (Lambertian)

Pixel intensity at position x,y in ith image.

surface radiance

$$I_i(x,y) = kL_i(x,y)$$

$$= k\rho(x,y)\hat{N}(x,y)\cdot\hat{S}_i$$

surface albedo

surface normal

ith light source direction

Combining all the measurements

$$\begin{pmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{pmatrix} = \begin{pmatrix} \hat{S}_1^T \\ \hat{S}_2^T \\ \vdots \\ \hat{S}_n^T \end{pmatrix} \rho(x,y)\hat{N}(x,y)$$

Solve for g(x,y). May be ill-conditioned

$$\begin{pmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{pmatrix} = \begin{pmatrix} \hat{S}_1^T \\ \hat{S}_2^T \\ \vdots \\ \hat{S}_n^T \end{pmatrix} \vec{g}(x,y)$$

A fix to avoid problems in dark areas: pre-multiply both sides by the image intensities

$$\begin{pmatrix} I_1(x,y) & 0 & \cdots & 0 \\ 0 & I_2(x,y) & \cdots & 0 \\ \vdots & \vdots & \vdots & 0 \\ 0 & 0 & 0 & I_n(x,y) \end{pmatrix} \begin{pmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{pmatrix}$$

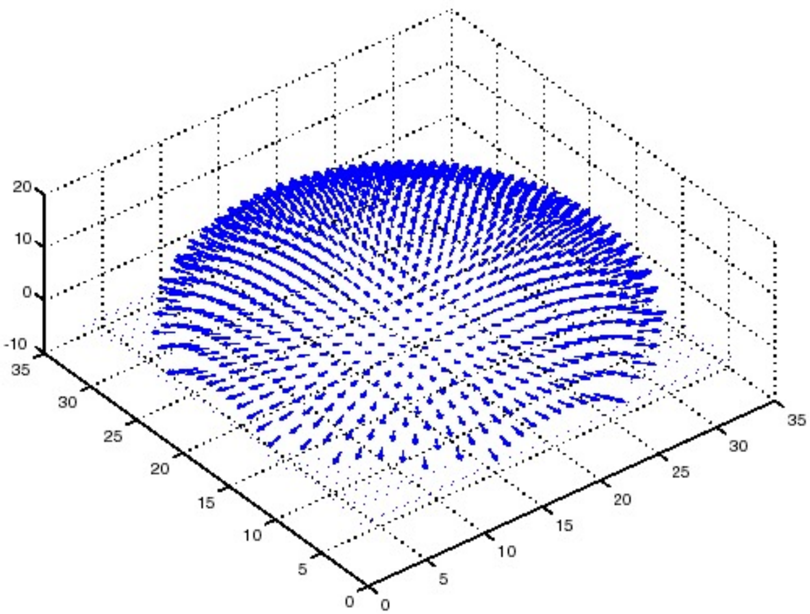
$$= \begin{pmatrix} I_1(x,y) & 0 & \cdots & 0 \\ 0 & I_2(x,y) & \cdots & 0 \\ \vdots & \vdots & \vdots & 0 \\ 0 & 0 & 0 & I_n(x,y) \end{pmatrix} \begin{pmatrix} \hat{S}_1^T \\ \hat{S}_2^T \\ \vdots \\ \hat{S}_n^T \end{pmatrix} \vec{g}(x,y)$$

Recovering albedo and surface normal

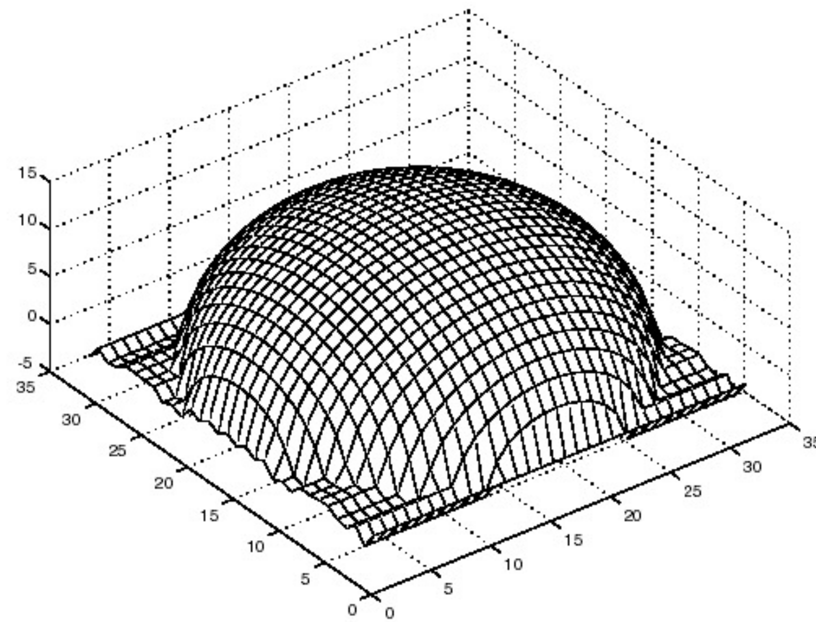
$$\rho(x, y) = \left|\vec{g}(x, y)\right|$$

$$\hat{N}(x, y) = \frac{\vec{g}(x, y)}{\left|\vec{g}(x, y)\right|}$$

Photometric Stereo

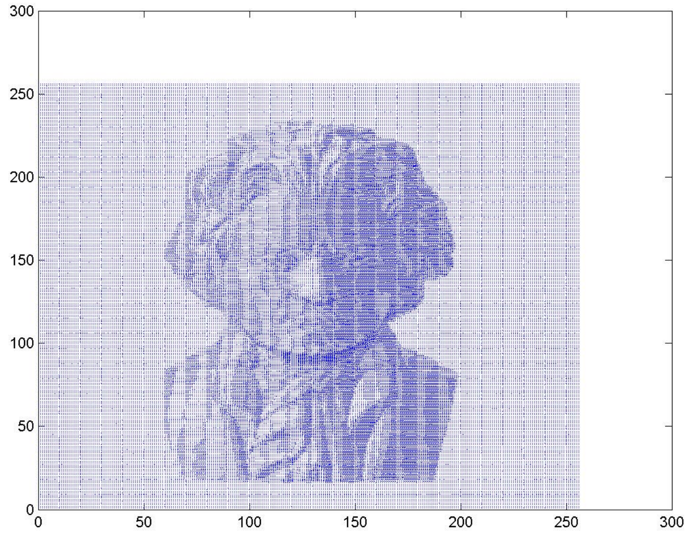


Surface normals



3D shape

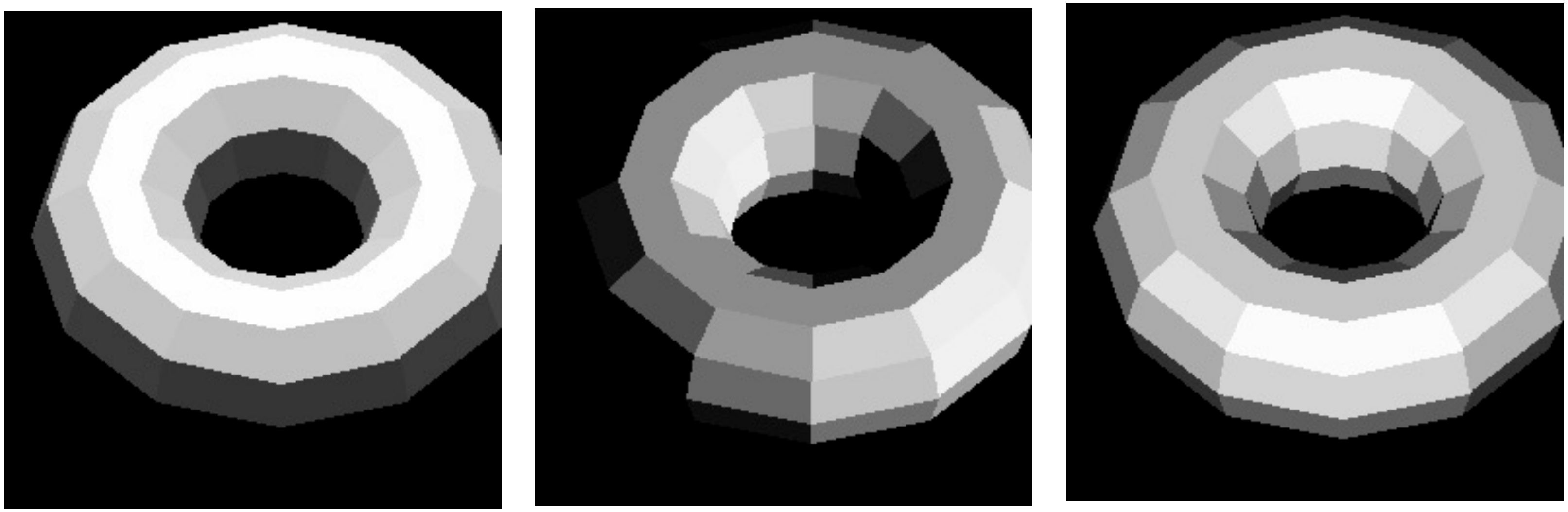
Photometric Stereo

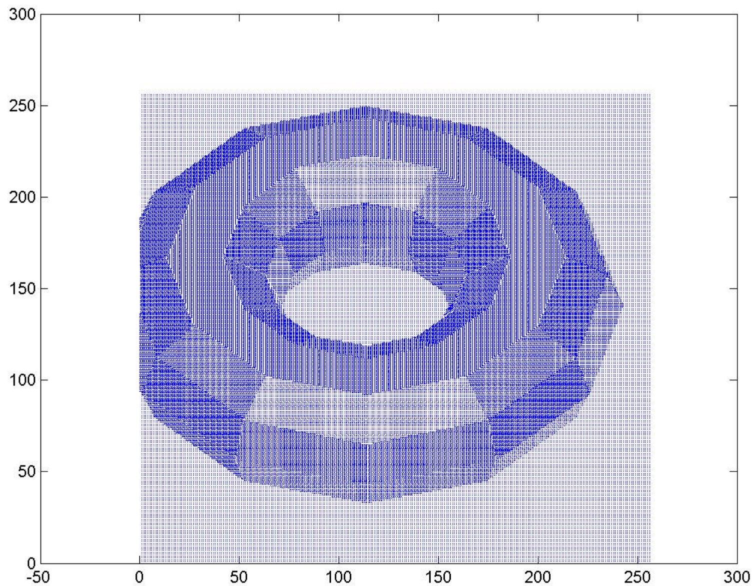


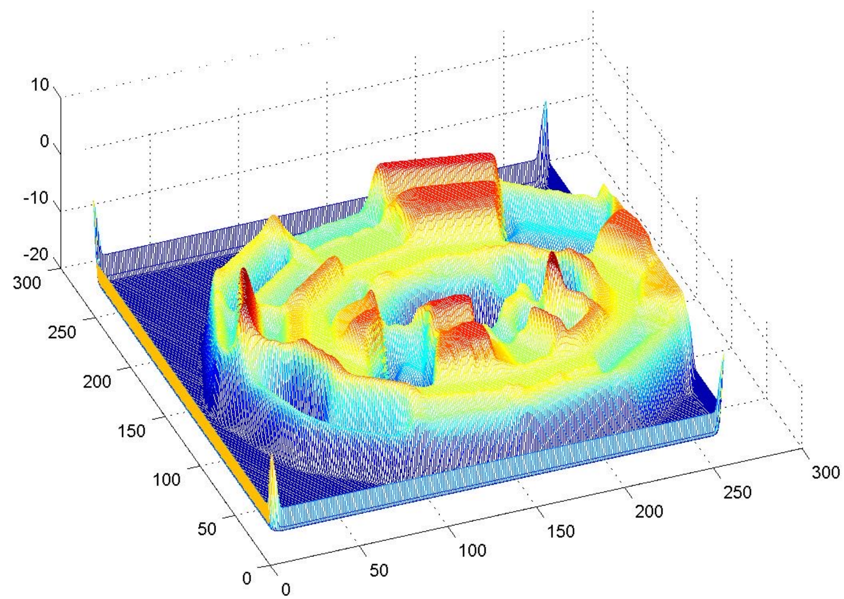
(by Xiaochun Cao)

Photometric Stereo

(by Xiaochun Cao)

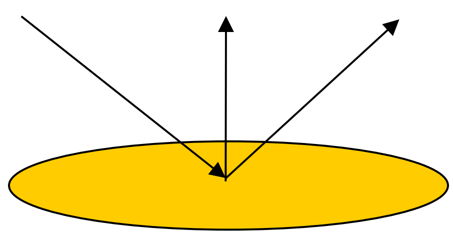






Specular Surface

- The glossy or mirror like surfaces are called *specular* surfaces.
- Radiation arriving along a particular direction can only leave along the *specular* direction, obtained from the surface normal.

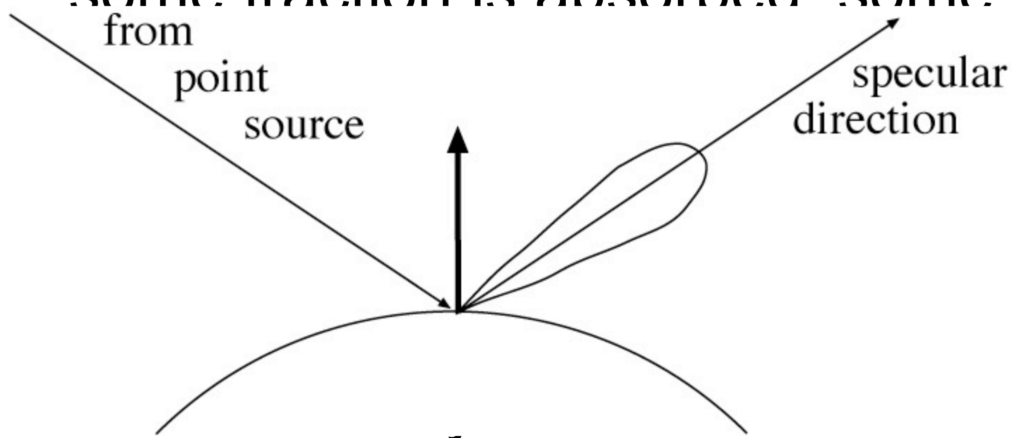


*The term *Specular* comes from the Latin word *speculum*, meaning mirror.

Specular surfaces

Another important class of surfaces is specular, or mirror-like.

- radiation arriving along a direction leaves along the specular direction
- reflect about normal
- some fraction is absorbed, some



Phong’s model

There are very few cases where the exact shape of the specular lobe matters.

Typically:

very, very small --- mirror

small -- blurry mirror

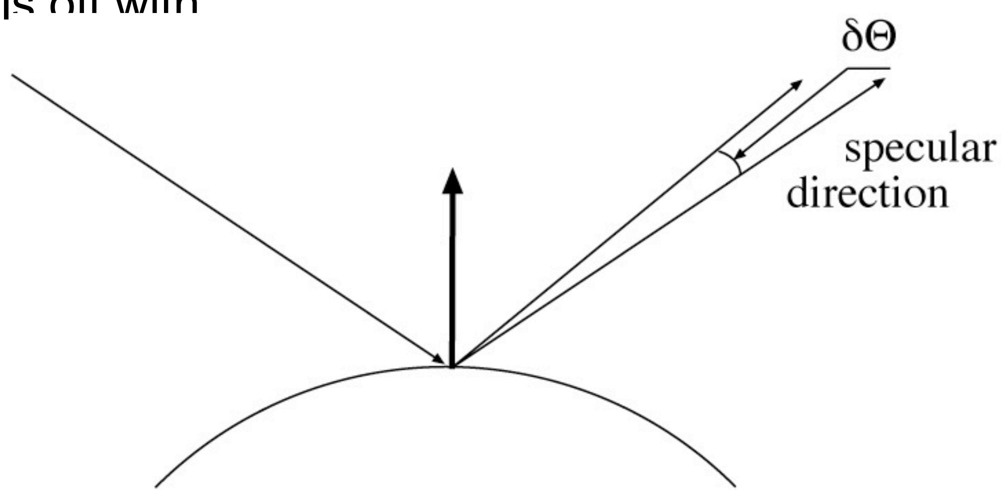
bigger -- see only light sources as “specularities”

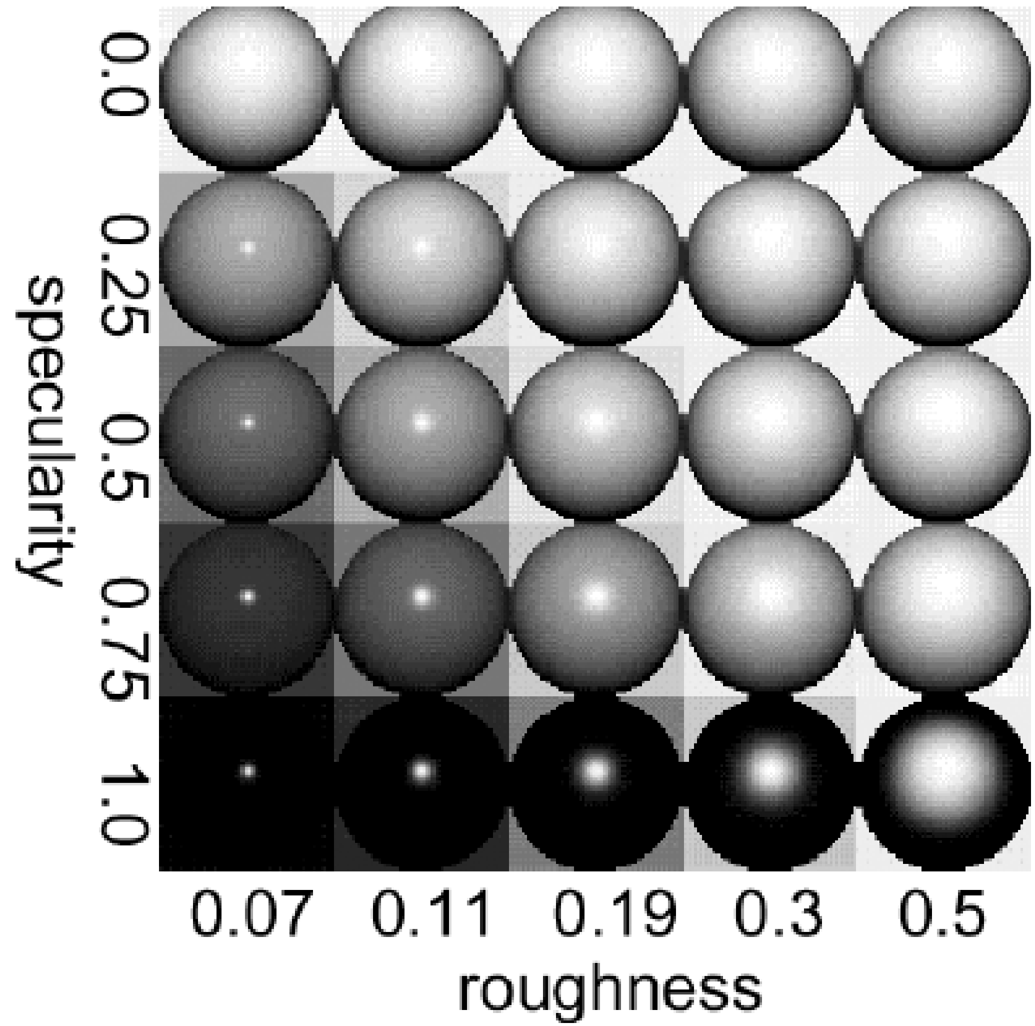
very big -- faint specularities

Phong’s model

reflected energy falls off with

$$\cos^{n}(\delta\vartheta)$$




0.0
0.25
0.5
0.75
1.0
specularity
0.07
0.11
0.19
0.3
0.5
roughness

Lambertian + specular model

Widespread model
- all surfaces are Lambertian plus specular component

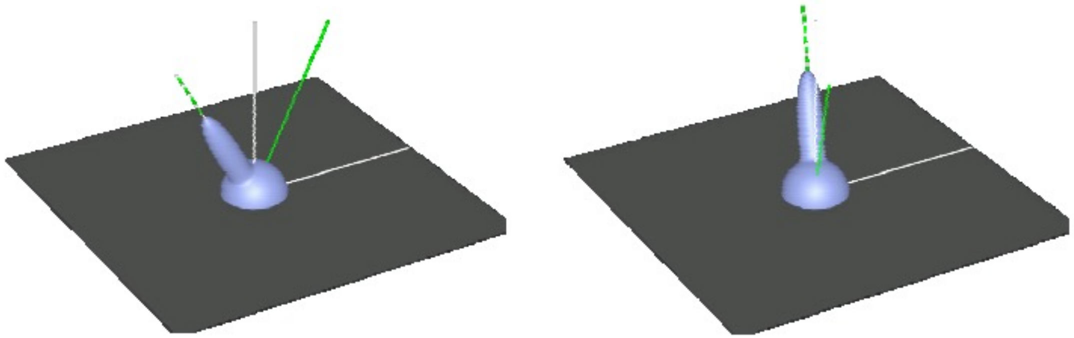
Advantages
- easy to manipulate
- very often quite close true

Disadvantages
- some surfaces are not
  - e.g. underside of CD's, feathers of many birds, blue spots on many marine crustaceans and fish, most rough surfaces, oil films (skin!), wet surfaces
- Generally, very little advantage in modelling behaviour of light at a surface in more detail -- it is quite difficult to understand behaviour of L+S surfaces

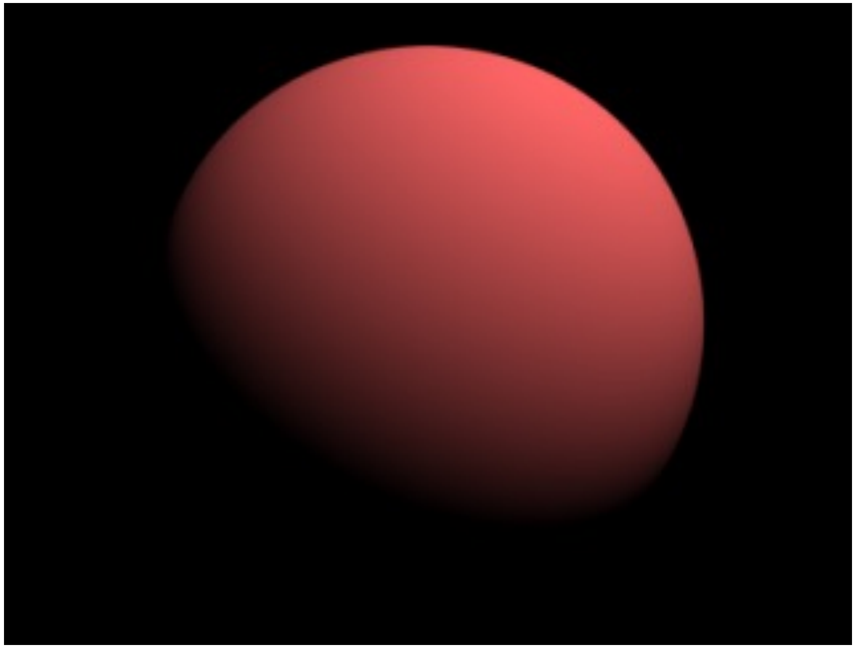
$$L(x,\theta_0,\phi_0) = \rho_l(x)\cos\theta_i + \rho_s(x)L(x,\theta_s,\phi_s)\cos^n(\theta_s-\theta_o)$$

Lambertian + Specular Model

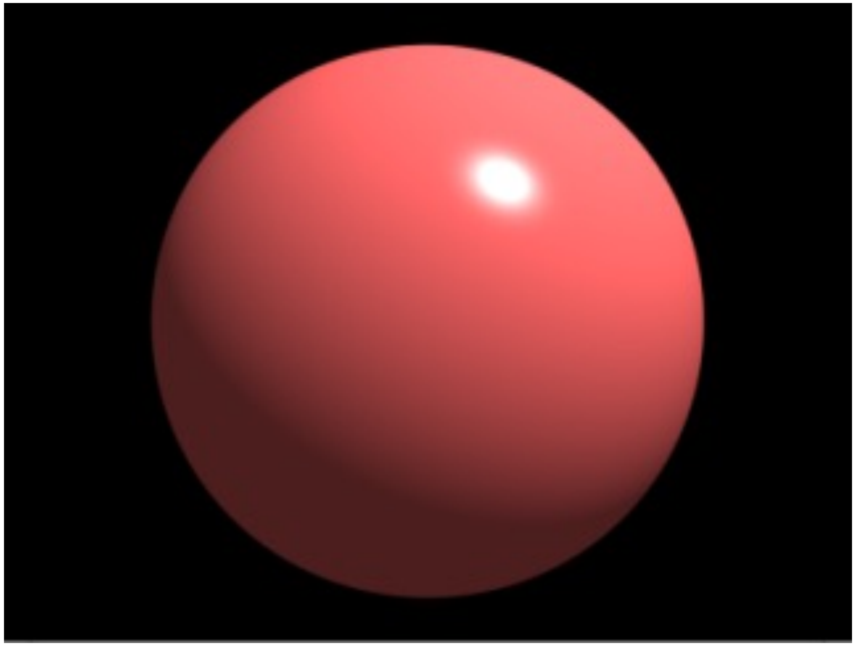
- Relatively few surfaces are either ideal diffuse or perfectly specular.
- The BRDF of many surfaces can be approximated as a combination of a Lambertian component and a specular component.



Lambertian + Specular Model



Lambertian



Lambertian + Specular